Clinical Study Summary Report (CSSR) Document Status: Final | **Version:** 1.0 | **Date:** 06-Apr-2026

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1. Administrative & Study Identification

Full Study Title: A Study of MK-1084 Plus Pembrolizumab (MK-3475) in Participants With KRAS G12C Mutant Non-small Cell Lung Cancer (NSCLC) With Programmed Cell Death Ligand 1 (PD-L1) Tumor Proportion Score (TPS) $\ge 50\%$ (MK-1084-004/KANDLELIT-004) **Official Regulatory Title:** A Phase 3, Randomized, Double-blind, Multicenter Study of MK-1084 in Combination With Pembrolizumab Compared With Pembrolizumab Plus Placebo as Firstline Treatment of Participants With KRAS G12C-Mutant, Locally Advanced or Metastatic NSCLC With PD-L1 TPS $\ge 50\%$ (KANDLELIT-004) **Short Title/Acronym:** MK-1084-004 / KANDLELIT-004 **Protocol Number:** MK-1084-004 **NCT Number:** NCT06345729 **Posting ID:** 681797069529240787 **Sponsor & Company Name:** Merck **Investigational Product:** Calderasib (MK-1084, a selective KRAS G12C inhibitor) + Pembrolizumab (MK-3475) **Phase of Development:** PHASE3 (Criteria Type: PHASE_REQUIREMENT) **Trial Status:** RECRUITING (Status Description: Currently recruiting participants) **Medical Monitor:** Merck Clinical Operations Lead

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2. Study Synopsis & Rationale

2.1 Study Objective Primary Hypotheses/Objectives: * **Hypothesis 1:** Combination of MK-1084 and pembrolizumab is superior to placebo plus pembrolizumab with respect to progression free survival (PFS) per Response Evaluation Criteria in Solid Tumors Version 1.1 (RECIST 1.1) by blinded independent central review (BICR). * **Hypothesis 2:** Combination of MK-1084 plus pembrolizumab is superior to placebo plus pembrolizumab with respect to overall survival (OS).

Secondary Objectives: To evaluate Objective Response Rate (ORR), Duration of Response (DoR), adverse events (AEs), treatment discontinuation due to AEs, and changes in quality of life.

2.2 Background & Rationale KRAS is among the most prevalent mutations in cancer, and KRAS G12C is the most common KRAS mutation in patients with non-small cell lung cancer (NSCLC). This study addresses the medical necessity of improving frontline therapeutic outcomes for previously untreated patients with KRAS G12C-mutated metastatic NSCLC whose tumors express PD-L1 (Tumor Proportion Score [TPS] $\ge 50\%$). Early evidence showed that combining the investigational oral selective KRAS G12C inhibitor MK-1084 with the PD-1 inhibitor pembrolizumab yielded promising anti-tumor activity and a manageable safety profile. This Phase 3 trial definitively compares the efficacy of this combination against standard-of-care pembrolizumab monotherapy.

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3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled, Active Comparator (Pembrolizumab) **Blinding:** Double-blind (Triple Masking: Participant, Investigator, Outcomes Assessor) **Randomization:** Randomized (Parallel Assignment)

3.2 Planned Enrollment & Duration Targeted Enrollment: 600 participants globally **Number of Sites:** Multicenter (Global) **Estimated Study Start:** 24-May-2024 **Estimated Primary Completion:** 19-Feb-2029

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4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Has a histologically or cytologically confirmed diagnosis of non-small cell lung cancer (NSCLC). 2. Has newly diagnosed Stage IIIB/IIIC NSCLC, not eligible for curative resection or curative chemotherapy/radiation as determined by a multidisciplinary tumor board and/or by radiation oncologist, surgeon, and medical oncologist or Stage IV (M1a, M1b, or M1c) by American Joint Committee on Cancer (AJCC) Staging Manual, Version 8. 3. Provides an archival tumor tissue sample (≤ 5 years) or newly obtained core, incisional, or excisional biopsy of a tumor lesion not previously irradiated to enable central laboratory testing of kirsten rat sarcoma (KRAS) G12C mutation status, PD-L1 status, and biomarker research. 4. If have had adverse events (AEs) due to previous anticancer therapies, must have recovered to < Grade 1 or baseline. 5. If human immunodeficiency virus (HIV)-infected, must have well controlled HIV on antiretroviral therapy (ART). 6. If Hepatitis B surface antigen (HBsAg) positive, have received Hepatitis B Virus (HBV) antiviral therapy for at least 4 weeks, and have undetectable HBV viral load. 7. If a participant has a history of Hepatitis C virus (HCV) infection, HCV viral load is undetectable.

4.2 Exclusion Criteria 1. Has diagnosis of small cell lung cancer. For mixed tumors, if small cell elements are present, the participant is ineligible. 2. Has active inflammatory bowel disease requiring immunosuppressive medication or previous clear history of inflammatory bowel disease. 3. Has known history of, or active, neurologic paraneoplastic syndrome. 4. Has an active infection requiring systemic therapy, with exceptions. 5. Has uncontrolled, significant cardiovascular disease or cerebrovascular disease. 6. Has one or more of the following ophthalmological findings/conditions: intraocular pressure >21 mmHg and/or any diagnosis of glaucoma, diagnosis of central serous retinopathy, retinal vein occlusion, or retinal artery occlusion, diagnosis of retinal degenerative disease. 7. Has received prior systemic anticancer therapy for their locally advanced or metastatic NSCLC. 8. Has received radiation therapy to the lung that is >30 Gray within 6 months of start of study intervention. 9. Has received radiotherapy within 2 weeks of start of study intervention. Participants must have recovered from all radiation-related toxicities, not required corticosteroids, and not have had radiation pneumonitis. 10. Has known active central nervous system metastases and/or carcinomatous meningitis. 11. Known additional malignancy that is progressing or has required active treatment within the past 3 years. 12. Has active autoimmune disease that has required systemic treatment in the past 2 years. 13. Has history of (noninfectious) pneumonitis/interstitial lung disease that required steroids or has current pneumonitis/interstitial lung disease. 14. Is HIV-infected and has a history of Kaposi's sarcoma and/or Multicentric Castleman's Disease. 15. Has history of allogenic tissue/solid organ transplant. 16. Has not fully recovered from any effects of major surgical procedure.

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5. Intervention & Treatment Plan

5.1 Investigational Regimen Investigational Product: Pembrolizumab (Drug Preferred Name: pembrolizumab, Trade Name: Keytruda, ATC Code: L01FF02) + MK-1084. EMA Reference: Pembrolizumab EPAR **Dosage/Frequency:** Pembrolizumab 200 mg administered on day 1 of each 21-day cycle (for up to 35 cycles) plus MK-1084 (calderasib) administered once daily. The control arm receives Pembrolizumab 200 mg IV Q3W plus matched placebo oral tablets once daily. **Route of Administration:** Oral (MK-1084 / Placebo) and Intravenous (Pembrolizumab). **Duration of Treatment:** Pembrolizumab up to 35 cycles (approximately 2 years); MK-1084/ Placebo continues until disease progression or unacceptable toxicity.

5.2 Concomitant & Prohibited Therapy Permitted: Supportive care medications as per investigator discretion and institutional standards. **Prohibited:** Other concurrent antineoplastic therapies, investigational agents, or treatments that strongly interact with KRAS G12C inhibition or PD-1 targeted immunotherapies.

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6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Tumor Tissue Testing (PD-L1 TPS & KRAS G12C), ECOG PS, Imaging (RECIST 1.1)
Baseline	Day 0	Randomization, First Dose of MK-1084/Placebo and Pembrolizumab
Interim	Q3W (21 Days)	Clinical safety assessments, Pembrolizumab IV infusion, Dispense Oral IP
End of Study	Follow-up	Final Safety Labs, Survival Follow-up, IP Accountability

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7. Outcome Measures (Endpoints)

7.1 Primary Endpoints 1. **Progression-Free Survival (PFS):** PFS is defined as the time from randomization until either documented disease progression per Response Criteria in Solid Tumors Version 1.1 (RECIST 1.1) or death due to any cause, whichever occurs first. PFS as determined by blinded independent central review (BICR) will be presented. 2. **Overall Survival (OS):** OS is defined as the time from randomization to death due to any cause.

7.2 Secondary Endpoints 1. Objective Response Rate (ORR). 2. Duration of Response (DOR). 3. Number of Participants Who Experience One or More Adverse Events (AEs).

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8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring of any-cause and treatment-related AEs via clinical assessment at each 21-day cycle visit. **Serious Adverse Events (SAEs):** Reported continuously per standard global pharmacovigilance and GCP timelines. **Data Monitoring Committee (DMC):** An independent Data Monitoring Committee will oversee the safety and efficacy data during the trial.

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9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for primary efficacy endpoints (PFS and OS). Safety Set for adverse event evaluations. **Sample Size Justification:** $N = 600$ targeted patients powered to detect a statistically significant superiority in PFS and OS for the MK-1084 + pembrolizumab arm compared to the placebo + pembrolizumab arm. **Handling of Missing Data:** Standard oncology trial statistical models (e.g., Kaplan-Meier estimates, Cox proportional hazards regression) with censoring for patients lost to follow-up or without events.

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10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine onsite and remote monitoring visits to assure protocol adherence, accurate data transcription, and safety reporting. **Audit Trail:** Secure EDC (Electronic Data Capture) systems with robust eCRF locking and query resolution processes.

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11. Study Identifiers & Governance

Primary ID: MK-1084-004 **NCT Number:** NCT06345729 **Posting ID:** 681797069529240787 **Source Level:** 5 **Sponsor/Lead Organization:** Merck **Study Title:** KANDLELIT-004

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12. Clinical Framework (Oncology Specific)

Disease Areas: Non-small Cell Lung Cancer **Disease Name:** Metastatic Non Small Cell Lung Cancer **Preferred Name:** Metastatic non-small cell lung cancer **Preferred UMLS Name:** KRAS G12C-Mutated Non-Small Cell Lung Carcinoma **Semantic Type:** Neoplastic Process **MeSH Code:** D016938 **SNOMED ID:** 457721000124104 **Diagnosis Threshold / Definition:** Stage IV includes: (Any T, Any N, M1a); (Any T, Any N, M1b). M1a: Lung cancer with separate tumor nodule(s) in a contralateral lobe; tumor with pleural nodules or malignant pleural (or pericardial) effusion. M1b: Distant metastasis. (AJCC 7th ed.) **Medical Methodology:** Targeted therapy combined with Immune Checkpoint Inhibition **Optimization Target:** Improvement in Progression-Free Survival (PFS) and Overall Survival (OS)

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13. Study Procedures & Methodology

Management Pathway: Standard oncological care pathway for locally advanced and metastatic NSCLC. **Education Protocol (HCP):** Site initiation visits and ongoing investigator meetings regarding the protocol and KRAS G12C testing requirements. **Education Protocol (Patient):** Informed consent process detailing the risks of combined targeted/immunotherapy. **Quality Audit Mechanism:** Centralized monitoring of tumor scans (BICR) and routine clinical quality assurance audits.

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14. Observation & Follow-Up Schedule

Total Duration: Until disease progression, unacceptable toxicity, or up to 35 cycles (approx. 2 years) for pembrolizumab, followed by survival follow-up. **Onsite Visit Cadence:** Every 3 weeks (Q3W) aligned with IV infusion cycles. **Remote Monitoring Frequency:** As per specific site monitoring plans. **Checklist Review Interval:** Ongoing AE and concomitant medication review prior to every cycle.

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15. Sign-off & Approval

| Role | Name | Signature | Date | | :--- | :--- | :--- | :--- |
| **Principal Investigator** | [Name] | ____ **| [DD-MMM-YYYY] | |**

Clinical Operations Lead | [Name] | ___ | *[DD-MMM-YYYY]* | | *Lead Statistician* | *[Name]* | _____ | [DD-MMM-YYYY] |